

VPN (Virtual Private Network)

✓ First Create Vnet- 1.subnet, 2.GatewaySubnet

Home > Network foundation | Virtual network

Create virtual network

Basics Security IP addresses Tags

Add a subnet

10.0.0.0/16

This address prefix overlaps with virtual network address space. [Learn more](#)

10.0.0.0

10.0.0.0 - 10.0.255.255

Subnets	IP addresses
Usubnet	10.0.0.0 -

Add a subnet

Select an address space and configure your subnet. You can customize a default sub

Subnet purpose ⓘ

Virtual Network Gateway

Name * ⓘ

GatewaySubnet

IPv4

Include an IPv4 address space

☒

IPv4 address range ⓘ

10.0.0.0/16

10.0.0.0 - 10.0.255.255

Starting address * ⓘ

10.0.1.0

Size ⓘ

/24 (256 addresses)

Subnet address range ⓘ

10.0.1.0 - 10.0.1.255

IPv6

Create virtual network

Basics Security IP addresses Tags Review + create

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure subscription 1

Resource group *

(New) UVnet-rg

[Create new](#)

Instance details

Virtual network name *

UVnet

Region * ⓘ

(US) Central US

[Deploy to an Azure Extended Zone](#)

Create virtual network ...

 Validation passed

- Basics
- Security
- IP addresses
- Tags
- Review + create**

Subscription	Azure subscription 1
Resource Group	UVnet-rg
Name	UVnet
Region	Central US

Security

Azure Bastion	Disabled
Azure Firewall	Disabled
Azure DDoS Network Protection	Disabled

IP addresses

Address space	10.0.0.0/16 (65,536 addresses)
Subnet	Usubnet (10.0.0.0/24) (256 addresses)
Subnet	GatewaySubnet (10.0.1.0/24) (256 addresses)

Home > UVnet-1771409408371 | Overview

 **UVnet**   ... 

Virtual network

- What is Azure Firewall Premium SKU
- Diagnose issues with this virtual network
- What are best practices for vNet design?

Search

- Move
- 🗑 Delete
- 🔄 Refresh
- 🗨 Give feedback

- Overview**
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
- Address space
- Connected devices
- Subnets

Essentials

Resource group (move)	: UVnet-rg	Address space	: 10.0.0.0/16
Location (move)	: Central US	Subnets	: 2 subnets
Subscription (move)	: Azure subscription 1	DNS servers	: Azure provided DNS service
Subscription ID	: 5d75b66d-66bf-44e0-8d7e-7e61e4b043d7	BGP community string	: Configure
		Virtual network ID	: 7b65e535-6dff-4f9b-b4d4-c36ecd4e7d32

Tags [\(edit\)](#) : [Add tags](#)

- Topology
- Properties
- Capabilities (5)**
- Recommendations
- Tutorials



✓ Next Search for **VNG** below

The screenshot shows the Microsoft Azure portal interface. At the top, the header includes the Microsoft Azure logo, an 'Upgrade' button, and a search bar containing the text 'virtual network gateway'. Below the header, the breadcrumb navigation shows 'Home > Hybrid connectivity'. The main heading is 'Hybrid connectivity | ExpressRoute gateways' with a 'Preview' tag. A search bar is present, and to its right are buttons for '+ Create', 'Manage view', and 'Refresh'. Below the search bar is a filter input field labeled 'Filter for any field...'. On the left, a sidebar lists navigation options: 'Overview', 'ExpressRoute' (expanded), 'Set up ExpressRoute', 'ExpressRoute circuits', 'ExpressRoute directs', and 'ExpressRoute gateways' (highlighted). On the right, a search results panel shows 'All', 'Services (63)', and 'Marketplace' tabs. Under the 'Services' tab, the following items are listed: 'Virtual network gateways', 'Virtual networks', 'Virtual machines', and 'Hybrid connectivity' (with keywords: 'Virtual network gateways'). Under the 'Marketplace' tab, 'Virtual network gateway' is listed.

✓ Select **VPN Gateway** below and **Create**

The screenshot shows the Microsoft Azure portal interface for 'Hybrid connectivity | VPN gateways'. The breadcrumb navigation shows 'Home > Hybrid connectivity'. The main heading is 'Hybrid connectivity | VPN gateways' with a 'Preview' tag. A search bar is present, and to its right are buttons for '+ Create', 'Manage view', and 'Refresh'. Below the search bar is a filter input field labeled 'Filter for any field...'. On the left, a sidebar lists navigation options: 'Overview', 'ExpressRoute', 'VPN gateway' (expanded), 'Set up VPN Gateway', 'VPN gateways' (highlighted), and 'VPN connections'. On the right, a dropdown menu is open from the '+ Create' button, showing the option 'Create new virtual network gateway'. Below this, a table lists existing VPN gateways with columns for a checkbox, 'Name', and an upward arrow. The table contains one entry: a checkbox, 'vVPN1', and an upward arrow.

Create virtual network gateway ...

Name *

UVpn01

✓

Region *

Central US

▼

[Deploy to an Azure Extended Zone](#)

Gateway Configuration

Gateway type *

☒ VPN

☐ ExpressRoute

SKU *

VpnGw1AZ

▼

Generation

Generation1

▼

Enable Advanced Connectivity

☐ Enabled

☒ Disabled

Virtual network *

UVnet

▼

[Create virtual network](#)

Subnet

GatewaySubnet (10.0.1.0/24)

▼

Only virtual networks in the currently selected subscription and region are listed.

Public IP address

Public IP address *

☒ Create new

☐ Use existing

Public IP address name *

UVPN-PIP

✓

Public IP address SKU

Standard

Assignment

☐ Dynamic

☒ Static

Enable active-active mode *

☐ Enabled

☒ Disabled

Review + create

Previous

Next : Tags >

[Download a template for automation](#)

Create virtual network gateway ...

✓ Validation passed

Subscription	Azure subscription 1
Resource group	UVnet-rg
Name	UVpn01
Region	Central US
SKU	VpnGw1AZ
Generation	Generation1
Virtual network	UVnet
Subnet	GatewaySubnet (10.0.1.0/24)
Gateway type	Vpn
VPN type	RouteBased
Enable active-active mode	Disabled
Enable Advanced Connectivity	Disabled
Configure BGP	Disabled
Public IP address	UVPN-PIPV

Create

Previous

Next

[Download a template for automation](#)

✓ Below VPG it will take atleast 30mins

Home



NoMarketplace-20260218155548 | Overview ...

Deployment

Search

× <<



Delete



Cancel



Redeploy



Download



Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress



Deployment name : NoMarketplace-202602181555... Start time : 18/02/2026, 16:02:15
Subscription : Azure subscription 1 Correlation ID : d55a6cd4-1024-4496-ae5a-a4e...
Resource group : UVnet-rg

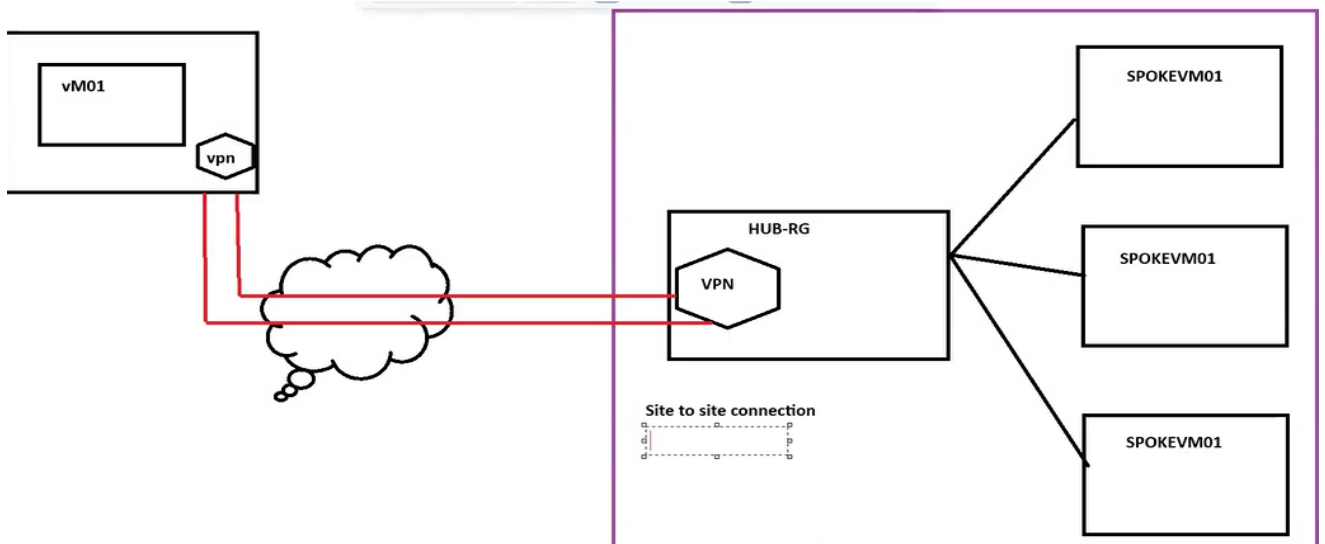
Deployment details

Resource	Type	Status	Operation
UVpn01	Virtual network gateway	Created	Operation
UVPN-PIPV	Public IP address	OK	Operation

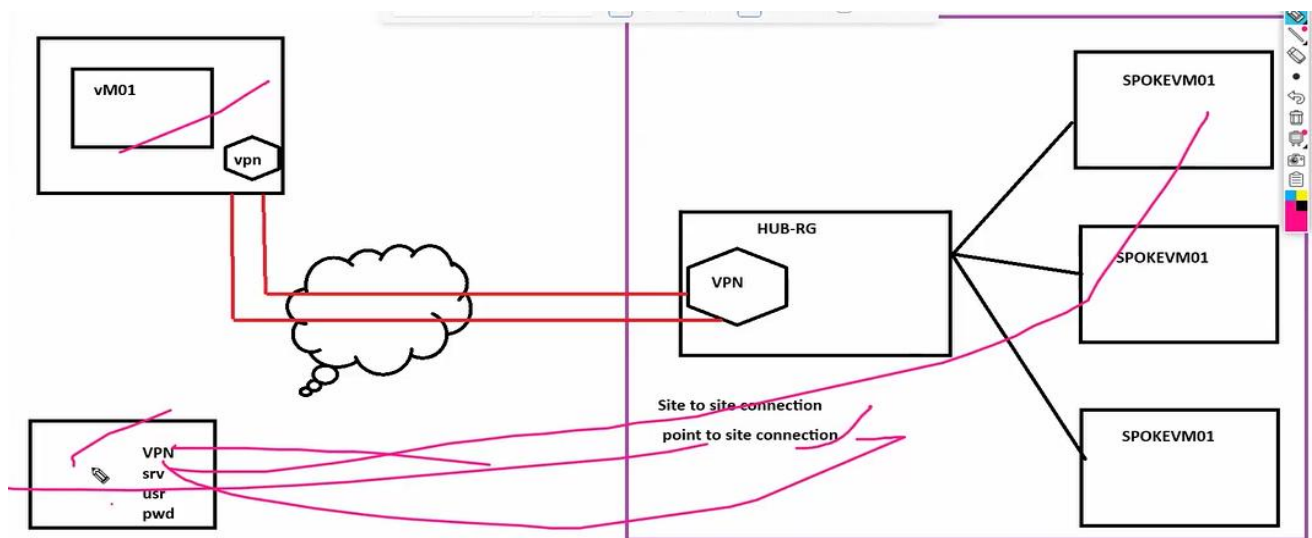
➤ To connect any machine most securely and privately we use VPN(VNG-VPG)

➤ Site-to-Site VPN → Network to Network

- ✓ When we create VPN, by default one tunnel is created and traffic will move through that Tunnel only.
- ✓ Traffic which comes from machines, it will travel through VPN from over the internet in encrypted mode and secured.



➤ Point-to-Site VPN → User to Network



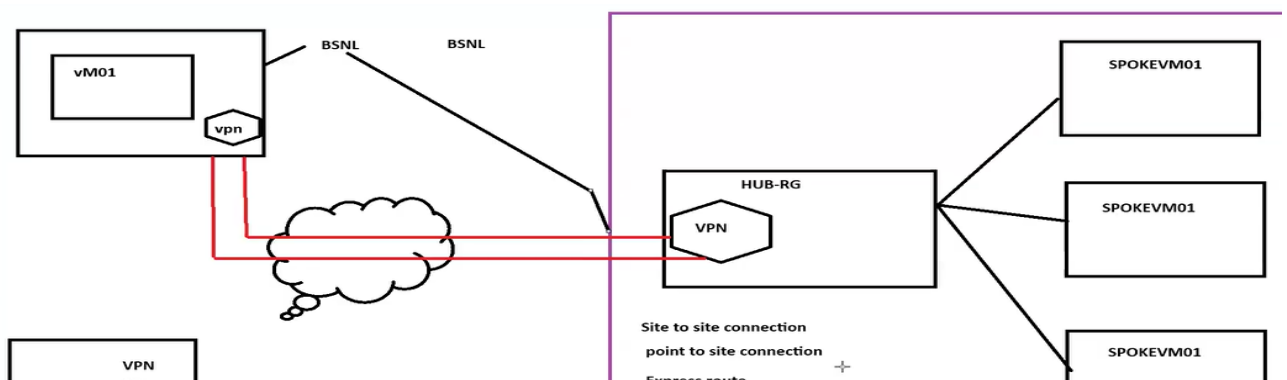
- Point-to-Site VPN allows **individual client devices** (laptop/desktop) to connect securely to a private network
- Connection is established **from a single user device to a VPN gateway**
- Commonly used for **remote employees / work-from-home**
- Works over the **public internet** but traffic is **encrypted**

NOTE: Above two connections are Over the Internet only

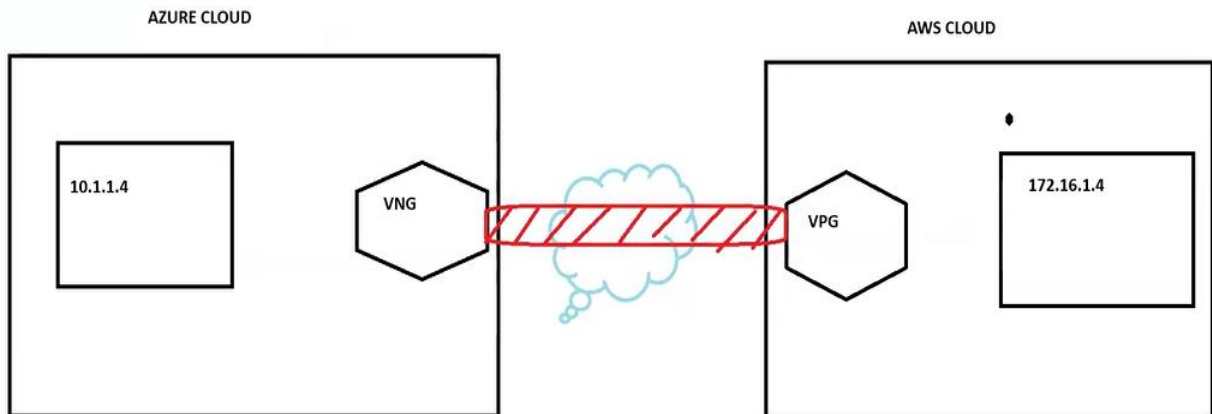
➤ Express Route

✓ **ExpressRoute = Private, fast, direct connection to Azure**

- ✓ ExpressRoute is a private, dedicated connection between on-premises data center and Microsoft Azure, not using the public internet.
- ✓ More **secure** than VPN
- ✓ **High speed, low latency**
- ✓ **Stable** for critical business apps



 For Lab- **SITE TO SITE CONNECTION** below



➤ AWS – First **create VPC** on AWS → **Search VPC** → **Create VPC**

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only

☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

UVPG-VPC01

IPv4 CIDR block [Info](#)

☒ IPv4 CIDR manual input

☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

172.16.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block

☐ IPAM-allocated IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

Tenancy [Info](#)

Default

VPC encryption control (\$) [Info](#)

Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#)

☒ None

☐ Monitor mode

See which resources in your VPC are unencrypted but allow the creation of unencrypted resources.

☐ Enforce mode

Requires all resources, except exclusions, in your VPC to be encryption-capable and blocks creation of unencrypted resources.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name



Value - optional

Q UVPG-VPC01



Remove tag

Add tag

You can add 49 more tags

Cancel

 Preview code

Create VPC

✓ Copy **VPC ID** below

VPC dashboard

AWS Global View

Filter by VPC:

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Edge-only Internet gateways

Carrier gateways

VPC option sets

You successfully created vpc-06bdafd2aa6f16392 / UVPG-VPC01

vpc-06bdafd2aa6f16392 / UVPG-VPC01

Details

VPC ID	State	Block Public Access	DNS hostnames
vpc-06bdafd2aa6f16392	Available	Off	Disabled
DNS resolution	Tenancy	DHCP option set	Main route table
Enabled	default	dopt-0c00429e7ae601485	rtb-0529d815b62846602
Main network ACL	Default VPC	IPv4 CIDR	IPv6 pool
acl-0cfa00403b8395a52	No	172.16.0.0/16	-
IPv6 CIDR (Network border group)	Network Address Usage metrics	Route 53 Resolver DNS Firewall rule groups	Owner ID
-	Disabled	-	220934192484

✓ Click on Subnets and search with VPC ID and create subnet.

Create subnet

VPC

VPC ID

Create subnets in this VPC.

vpc-0d37c9c72c1fcb24d (UVPG-VPC01)

Associated VPC CIDRs

IPv4 CIDRs

172.16.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

UVPGsubnet01

The name can be up to 256 characters long.

Availability Zone

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

172.16.0.0/16

IPv4 subnet CIDR block

172.16.1.0/24

VPC dashboard <

AWS Global View

Filter by VPC: ▼

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

Subnets (1) [Info](#)

Last updated 2 minutes ago [Action](#)

Find subnets by attribute or tag

vpc-0d37c9c72c1fcb24d ✕ [Clear filters](#)

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	UVPGsubnet01	subnet-024de9ddafb7ea51e	✔ Available	vpc-0d37c9c72c1fcb24d UVP...	⊖ Off	172.16.1.0/24

AWS --- VPC subnet Internet Gateway	AZURE Resource group <u>Vnet</u> Subnet ---
--	---

✓ Created subnet above

✓ let's create Internet Gateway and attach to VPC

≡ [VPC](#) > [Internet gateways](#) > Create internet gateway

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

UVPG-IGW

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q UVPG-IGW

[Remove](#)

[Add new tag](#)

You can add 49 more tags.

[Cancel](#)

[Create internet gateway](#)

✓ Click below Attach to VPC

✓ The following internet gateway was created: igw-0c27d45497070b704 - UVPG-IGW. You can now attach to a VPC to enable the VPC to communicate with the internet. [Attach to a VPC](#) ✕

igw-0c27d45497070b704 / UVPG-IGW [Actions](#)

Details [Info](#)

Internet gateway ID	State	VPC ID	Owner
igw-0c27d45497070b704	Detached	-	220934192484

Tags (1) [Manage tags](#)

Key	Value
Name	UVPG-IGW

Attach to VPC (igw-0c27d45497070b704) [Info](#)

VPC
 Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
 Attach the internet gateway to this VPC.

✕

▶ AWS Command Line Interface command

[Cancel](#) [Attach internet gateway](#)

- ✓ **To connect to the internet, we need internet gateway in AWS**
- ✓ **And route table will suggest to IGW to connect.**

AWS	AZURE
---	Resource group
VPC	Vnet
subnet	Subnet
Internet Gateway	---
route tables	---

- ✓ **Copy vpc id and search on route tables and edit route to internet gateway**
- ✓ **Edit Routes below**

VPC dashboard < **Route tables**

AWS Global View 

Filter by VPC: ▼

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Route tables (1/1) [Info](#)

Last updated 2 minutes ago [Actions](#) [Create route table](#)

Find route tables by attribute or tag

VPC: vpc-0d37c9c72c1fcb24d ✕ [Clear filters](#)

☒	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☒	UVPG-RT 🔗	rtb-0a133c64507b90521	-	-	Yes	vpc-0d37c9c72c1fcb24d UVP...	220934192484

rtb-0a133c64507b90521 / UVPG-RT

[Details](#) [Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

Routes (1) Both [Edit routes](#)

Filter routes

Destination	Target	Status	Propagated	Route Origin
172.16.0.0/16	local	Active	No	Create Route Table

✓ Choose any and add target → internet gateway ID → Save

Edit routes

Destination	Target	Status	Propagated	Route Origin
172.16.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

[Add route](#)

[Cancel](#) [Preview](#) [Save changes](#)

VPC dashboard < **Route tables** > rtb-0a133c64507b90521

AWS Global View 

Filter by VPC: ▼

Virtual private cloud

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Internet gateways

Egress-only Internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Updated routes for rtb-0a133c64507b90521 / UVPG-RT successfully Details ✕

rtb-0a133c64507b90521 / UVPG-RT [Actions](#)

Details [Info](#)

Route table ID rtb-0a133c64507b90521	Main Yes	Explicit subnet associations -	Edge associations -
VPC vpc-0d37c9c72c1fcb24d UVPG-VPC01	Owner ID 220934192484		

[Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

Routes (2) Both [Edit routes](#)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0c27d45497070b704	Active	No	Create Route
172.16.0.0/16	local	Active	No	Create Route Table

✓ Next copy VPC ID → go to Security Group → search with VPC id

The screenshot shows the AWS Management Console 'Security Groups' page. A search filter 'vpc-0d37c9c72c1fcb24d' is applied. A table lists security groups, with the 'default' group selected. An 'Edit Name' dialog box is open, showing the current name 'UVPC-SG' and buttons for 'Cancel' and 'Save'.

Name	Security group ID	Security group name	VPC ID	Description
default	sg-0a40fac299a5f654a	default	vpc-0d37c9c72c1fcb24d	default VPC security group

✓ Click on inbound rules and edit it below

The screenshot shows the 'Inbound rules' tab for the 'sg-0a40fac299a5f654a' security group. It displays a table with one inbound rule: 'sgr-06f8f5cceb9dd8824' for 'All traffic' on 'All' ports.

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-06f8f5cceb9dd8824	-	All traffic	All	All	sg-

✓ Delete default rule and click on Add rule and select below

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID

Type [Info](#)Protocol [Info](#)Port range [Info](#)Source [Info](#)Description - optional [Info](#)

-

Custom TCP

TCP

22

Anywh...

Q

Delete

Add rule

Custom

0.0.0.0/0 X

Anywhere-IPv4 ✓

Anywhere-IPv6

My IP

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security groups from known IP addresses only.

Cancel

Preview changes

Save rules

Egress-only Internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

Security

Network ACLs
Security groups

PrivateLink and Lattice

Getting started
Endpoints
Endpoint services
Service networks
Lattice services
Resource configurations
Resource gateways

Inbound security group rules successfully modified on security group (sg-0a40fac299a5f654a | default)

Details

Security Groups (1/2) [Info](#)

Actions

Export security groups to CSV

Create security group

Find security groups by attribute or tag

	Name	Security group ID	Security group name	VPC ID	Description
☒	UVPC-SG	sg-0a40fac299a5f654a	default	vpc-0d37c9c72c1fcb24d	default VPC security group
☐	-	sg-0a56168bf96f9c383	default	vpc-04908860bf513794f	default VPC security group

sg-0a40fac299a5f654a - default

Details

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-		sgr-090ee3e868174d010	IPv4	SSH	TCP	22	0.0.0.0/0

➤ Next Create EC2

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instances [Info](#)Last updated
less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
--	------	-------------	----------------	---------------	--------------	--------------	-------------------	-----------------

No instances

You do not have any instances in this region

Launch instances

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

UEc2-01

Add additional tags

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Q

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-019715e0d74f695be (64-bit (x86)) / ami-0848881f2a3dcebd1 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture

64-bit (x86) ▼

AMI ID

ami-019715e0d74f695be

Publish Date

2025-12-12

Username ⓘ

ubuntu

Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0112 USD per Hour
On-Demand SUSE base pricing: 0.0112 USD per Hour On-Demand Windows base pricing: 0.0204 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour

Free tier eligible

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Udayawskey

[Create new key pair](#)

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-0d37c9c72c1fcb24d (UVPG-VPC01)
172.16.0.0/16

[Create new VPC](#)

Subnet [Info](#)

subnet-024de9dda7b7ea51e UVPGsubnet01
VPC: vpc-0d37c9c72c1fcb24d Owner: 220934192484 Availability Zone: ap-south-1a (aps1-az1)
Zone type: Availability Zone IP addresses available: 251 CIDR: 172.16.1.0/24

[Create new subnet](#)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups [Info](#)

Select security groups

default sg-0a40fac299a5f654a
VPC: vpc-0d37c9c72c1fcb24d

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

✓ Created EC2 successfully

[EC2](#) > [Instances](#) > [Launch an instance](#)



Success

Successfully initiated launch of instance ([i-05254154c0182b8f0](#))

[VPC](#) > Virtual private gateways

groups

VPC endpoint associations

Virtual private network (VPN)

Customer gateways

Virtual private gateways

Site-to-site VPN concentrators

Site-to-Site VPN connections

Virtual private gateways [Info](#)

Find resource by attribute or tag

Name	Virtual private gateway ID	State	VPC attachment state	Type	VPC	Amazon A
------	----------------------------	-------	----------------------	------	-----	----------

No virtual private gateways

You do not have any virtual private gateways in this region

[Create virtual private gateway](#)

Create virtual private gateway [Info](#)

A virtual private gateway is the VPN Concentrator on the Amazon side of the site-to-site VPN connection.

Details

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

UVPGaws-azure

Value must be 256 characters or less in length.

Autonomous System Number (ASN)

☒ Amazon default ASN

☐ Custom ASN

Tags - optional

Use tags to label your AWS resources and search, filter, or track costs. Each tag consists of a key and an optional value. We recommend using a "Name" key tag to more easily identify your resources.

Key

Q Name

Value - optional

Q UVPGaws-azure

Remove

Add new tag

You can add up to 49 more tags.

Cancel

Create virtual private gateway

✓ Attach vpg to vpc below

🟢 You successfully created vgw-04e468e4ab3716def / UVPGaws-azure.

Virtual private gateways (1/1) [Info](#)

Q Find resource by attribute or tag

Name	Virtual private gateway ID	State	VPC attachment state	Type	
UVPGaws-azure	vgw-04e468e4ab3716def	Available	Detached	ipsec.1	Actions Attach to VPC Detach from VPC Manage tags Delete virtual private gateway

✓ Click attach to vpc below

Attach to VPC [Info](#)

Attach the virtual private gateway to this VPC.

Details

Virtual private gateway ID

vgw-04e468e4ab3716def

Available VPCs

Attach the virtual private gateway to this VPC.

Select a VPC by ID or Name

Q

vpc-0d37c9c72c1fcb24d / UVPG-VPC01

vpc-04908860bf513794f

Cancel

Attach to VPC

VPC > Customer gateways

configurations
Network Firewall resource groups
VPC endpoint associations

▼ Virtual private network (VPN)
Customer gateways
Virtual private gateways

Customer gateways Info

Find resource by attribute or tag

Name	Customer gateway ID	State	BGP ASN	IP address	Type	Certificate ARN
No customer gateways						
You do not have any customer gateways in this region						

Create customer gateway

✓ Copy azure gateway public IP and paste in below IP address.

Create customer gateway Info

A customer gateway is a resource that you create in AWS that represents the customer gateway device in your on-premises network.

Details

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

AWS-customer gateway

Value must be 256 characters or less in length.

BGP ASN Info

The ASN of your customer gateway device.

65000

Value must be in 1 - 4294967294 range.

IP address Info

Specify the IP address for your customer gateway device's external interface.

4.249.248.6

Certificate ARN - optional

The ARN of a private certificate provisioned in AWS Certificate Manager (ACM).

Select certificate ARN

Device - optional

Enter a name for the customer gateway device.

Enter device name

VPC > Customer gateways

configurations
Network Firewall resource groups
VPC endpoint associations

▼ Virtual private network (VPN)
Customer gateways
Virtual private gateways

You successfully created cgw-05a8dde9abc1dd43b / AWS-customer gateway.

Customer gateways (1) Info

Find resource by attribute or tag

Name	Customer gateway ID	State	BGP ASN	IP address	Type	Certificate ARN
AWS-customer gate...	cgw-05a8dde9abc1dd43b	Available	65000	4.249.248.6	ipsec.1	-

VPC > VPN connections

configurations
Network Firewall resource groups
VPC endpoint associations

▼ Virtual private network (VPN)
Customer gateways
Virtual private gateways
Site-to-site VPN concentrators
Site-to-Site VPN connections

VPN connections Info

Find resource by attribute or tag

Name	VPN ID	State	Virtual private gateway	Transit gateway	VPN Concentrator ID	Customer gateway
No VPN connections						
You do not have any VPN connections in this region						

Create VPN connection

- ✓ Below select **static IP** and give **AZURE Range 10.0.0.0/16-** below

Create VPN connection [Info](#)

Select the resources and additional configuration options that you want to use for the site-to-site VPN connection.

Details

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

Value must be 256 characters or less in length.

Target gateway type [Info](#)

- ☒ Virtual private gateway
☐ Transit gateway
☐ Site-to-Site VPN Concentrator
☐ Not associated

Virtual private gateway

Customer gateway [Info](#)

- ☒ Existing
☐ New

Customer gateway ID

Routing options [Info](#)

- ☐ Dynamic (requires BGP)
☒ Static

Static IP prefixes [Info](#)

VPC > VPN connections

Network Firewall rule groups

TLS inspection configurations

Network Firewall resource groups

VPC endpoint associations

Virtual private network (VPN)

Customer gateways

Virtual private gateways

Site-to-site VPN concentrators

[Site-to-Site VPN connections](#)

✓ You successfully created vpn-0fa6df68517c452ff / AWS-VPN-connection.

VPN connections (1) [Info](#)

	Name	VPN ID	State	Virtual private gateway	Transit gateway	VPN Concentrator ID
☐	AWS-VPN-connection	vpn-0fa6df68517c452ff	Pending	vgw-04e468e4ab3716def	-	-

Select a VPN connection

- ✓ Here go to routes and specify Azure Vnet (privateIP) range to VPG-see below

Edit routes

Destination	Target	Status
172.16.0.0/16	local	Active
	local	
10.0.0.0/16	Virtual Private Gateway	Active
	vgw-006c08c94999ebdf0	
0.0.0.0/0	Internet Gateway	Active
	igw-0df468866fe513909	

[Add route](#)

✓ Download below configuration

VPC > VPN connections

You successfully created vpn-0fa6df68517c452ff / AWS-VPN-connection.

VPN connections (1/1) Info

Find resource by attribute or tag

Name	VPN ID	State	Virtual private gateway	Transit gateway	VPN Concentrator ID
AWS-VPN-connection	vpn-0fa6df68517c452ff	Pending	vgw-04e468e4ab3716def	-	-

Actions Download configuration Create VPN connection

✓ Go to VPC route tables and edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
172.16.0.0/16	local	Active	No	CreateRouteTable
	local			
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute
	igw-0c27d45497070b704			
10.0.0.0/16	Virtual Private Gateway	-	No	CreateRoute
	vgw-04e468e4ab3716def			

[Add route](#)

Cancel Preview [Save changes](#)

Connections (1/1) Info

source by attribute or tag

VPN ID

vpn-0fa6df68517c452ff

VPN connection

vpn-0fa6df68517c452ff

Tunnel details

Static routing

Files

vpn-0fa6df68517c452ff

gateway

Download configuration

Download a sample configuration based on your customer gateway. Note that this is a sample only, and that it will require modification for using Advanced Algorithms, Certificates, and/IPv6.

Vendor
The manufacturer of the customer gateway device (for example, Cisco Systems, Inc).

Cisco Systems, Inc.

Platform
The class of the customer gateway device (for example, J-Series).

ASA 5500 Series

Software
The operating system running on the customer gateway device (for example, ScreenOS).

ASA 8.2+

IKE version
The IKE version you are using for your VPN connection.

ikev1

Include sample type - optional

☐ Enable

Sample type

Cancel

Download

Download configuration

Create VPN connection

VPN Concentrator ID

cgw-05a8dde9abc1dd43b

Customer gateway

cgw-05a8dde9abc1dd43b

Category

VPN

Authentication

! If not behind NAT, and you are not using an Accelerated VPN, we recommend disabling NAT-T. If you are using an Accelerated VPN, make sure that NAT-T is enabled.

```
!
crypto isakmp identity address
crypto isakmp enable <outside_interface>
crypto isakmp policy 201
  encryption aes
  authentication pre-share
  group 2
  lifetime 28800
  hash sha
exit
```

```
!
! The tunnel group sets the Pre Shared Key used to authenticate the
! tunnel endpoints.
```

```
!
tunnel-group 3.6.191.41 type ipsec-l2l
tunnel-group 3.6.191.41 ipsec-attributes
  pre-shared-key KZ_0xNMxLYRE4yFDdhZtMK_53nBypuD.
```

```
!
! This option enables IPsec Dead Peer Detection, which causes semi-periodic
! messages to be sent to ensure a Security Association remains operational.
```

```
!
  isakmp keepalive threshold 10 retry 10
exit
!
tunnel-group 3.109.222.112 type ipsec-l2l
tunnel-group 3.109.222.112 ipsec-attributes
  pre-shared-key 8TT7_nR0tXY2xHP1KOGa68HOGPKFUm7
```

Ln 64, Col 25 | 10,900 characters | Plain text | 100% | Unix (LF) | UTF-8

➤ Next go to Local network gateway below

Home



Hybrid connectivity | Local network gateways

Preview



Recommend connectivity for lis

Search

+ Create



Manage view



Refresh



Export to CSV



Open query



Overview

> ExpressRoute

✓ VPN gateway



Set up VPN Gateway



VPN gateways



VPN connections



Local network gateways

> Virtual WAN

Filter for any field...

Subscription equals all

Resource Group equals all



No local network gateway

Create a local network gateway to represent the on-premises site. A local network gateway specifies the public IP address of the VPN premises site. Later, create a VPN gateway connection between the

Microsoft Azure

Search resources, services, and docs (G+/)

Home > Hybrid connectivity | Local network gateways

Create local network gateway

Subscription * Azure subscription 1

Resource group * UVnet-rg
[Create new](#)

Instance details

Region * Central US

Name * localnetworkazure

Endpoint ⓘ IP address FQDN

IP address * ⓘ

Address Space(s) ⓘ

Add additional address range

[Review + create](#) [Previous](#) [Next: Advanced >](#)

```
File Edit View H1 ⌵ ⌵ B I ⌵ ...  
authentication pre-share  
group 2  
lifetime 28800  
hash sha  
exit  
!  
! The tunnel group sets the Pre Shared Key used to authenticate the  
! tunnel endpoints.  
!  
tunnel-group 3.6.191.41 type ipsec-l2l  
tunnel-group 3.6.191.41 ipsec-attributes  
pre-shared-key KZ_0xNMxLYRE4yfDdhZtMK_53nBypuD.  
!  
! This option enables IPsec Dead Peer Detection, which causes semi-perio  
! messages to be sent to ensure a Security Association remains operation  
!  
isakmp keepalive threshold 10 retry 10  
exit  
!  
tunnel-group 3.109.222.112 type ipsec-l2l  
tunnel-group 3.109.222.112 ipsec-attributes  
pre-shared-key 9TTZ.pr0tXX2xHRJKQGe68HQGPKFWumJ  
!  
! This option enables IPsec Dead Peer Detection, which causes semi-perio  
! messages to be sent to ensure a Security Association remains operation  
!  
isakmp keepalive threshold 10 retry 10  
exit  
!  
Ln 64, Col 24 | 10 of 10,900 charac | Plain text | 100% | Unix (LF) | UTF-8
```

Create local network gateway ...

Subscription *

Azure subscription 1

Resource group *

UVnet-rg

Create new

Instance details

Region *

Central US

Name *

localnetworkazure

Endpoint ⓘ

IP address

FQDN

IP address * ⓘ

3.6.191.41

Address Space(s) ⓘ

172.16.0.0/16

Add additional address range

Review + create

Previous

Next : Advanced >

Home > Hybrid connectivity | VPN gateways > UVpn01 | Connections

Create connection ...

Create a secure connection to your virtual network by using VPN Gateway or ExpressRoute.
[Learn more about VPN Gateway](#)
[Learn more about ExpressRoute](#)

Project details

Subscription *

Azure subscription 1

Resource group *

UVnet-rg

Create new

Instance details

Connection type * ⓘ

Site-to-site (IPsec)

Name *

AZURE-AWS

Region *

Central US

Previous | Next : Settings > |

portal.azure.com/?l=en-en-gb#view/Microsoft_Azure_HybridNetworking/

Microsoft Azure Search resources, services, and docs (G+)

Home > Hybrid connectivity | VPN gateways > UVpn01 | Connections

Create connection

Local network gateway * ⓘ

Authentication Method ⓘ

Shared Key(PSK) * ⓘ

IKE Protocol ⓘ

Use Azure Private IP Address ⓘ

Enable BGP ⓘ

IPsec / IKE policy ⓘ

Use policy based traffic selector ⓘ

DPD timeout in seconds * ⓘ

localnetworkazure

☒ Shared Key(PSK) ☐ Key Vault Certificate (Preview)

.....

☐ IKEv1 ☒ IKEv2

☒

☐

Default Custom

Enable Disable

45

Previous | Next : Settings > |

! The tunnel group sets the Pre Shared Key used to authenticate the
! tunnel endpoints.

!

```
tunnel-group 3.6.191.41 type ipsec-l2l
```

```
tunnel-group 3.6.191.41 ipsec-attributes
```

```
pre-shared-key KZ_0xNMXYRE4yfDdhZtMK_53nBypuD.
```

!

! This option enables IPsec Dead Peer Detection, which causes semi-periodic
! messages to be sent to ensure a Security Association remains operational.

!

```
isakmp keepalive threshold 10 retry 10
```

```
exit
```

!

```
tunnel-group 3.109.222.112 type ipsec-l2l
```

```
tunnel-group 3.109.222.112 ipsec-attributes
```

```
pre-shared-key 9TTZ.pr0tXX2xHRJKOG68HOGPKFWumJ
```

✓ Copy above key and paste below

Create connection ...

Virtual network gateway

To use a virtual network with a connection, it must be associated to a virtual network gateway.

Virtual network gateway *	①	UVpn01	▼
Local network gateway *	①	localnetworkazure	▼
Authentication Method	①	☒ Shared Key(PSK) ☐ Key Vault Certificate (Preview)	
Shared Key(PSK) *	①	 ✓	
IKE Protocol	①	☐ IKEv1 ☒ IKEv2	
Use Azure Private IP Address	①	☐	
Enable BGP	①	☐	
IPsec / IKE policy	①	<input checked="" type="button"/> Default Custom	
Use policy based traffic selector	①	Enable <input checked="" type="button"/> Disable	
DPD timeout in seconds *	①	45	
Connection Mode	①	☒ Default ☐ InitiatorOnly ☐ ResponderOnly	

Previous

Next : Tags >

Create connection ...

✓ Validation passed

Basics Settings Tags Review + create

Basics

Subscription	Azure subscription 1
Resource Group	UVnet-rg
Region	Central US
Connection type	Site-to-site (IPsec)
Connection name	AZURE-AWS

Settings

Virtual network gateway	UVpn01
Local network gateway	localnetworkazure
IKE Protocol	IKEv2
IPsec / IKE policy	Default
Use policy based traffic selector	Disable
DPD timeout in seconds	45
Connection Mode	Default
Shared Key(PSK)	9TTZ.pr0tXX2xHRJKQGe68HQGPKFWumJ

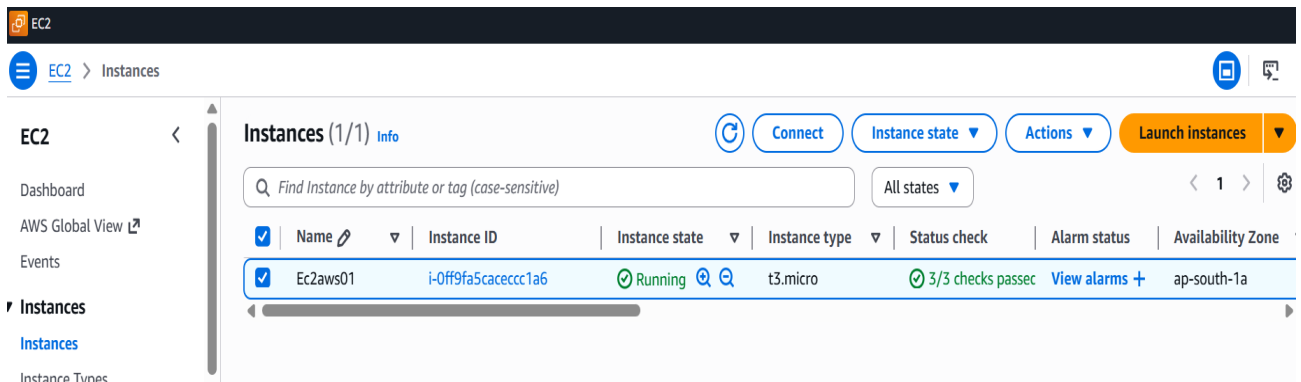
Create

Previous

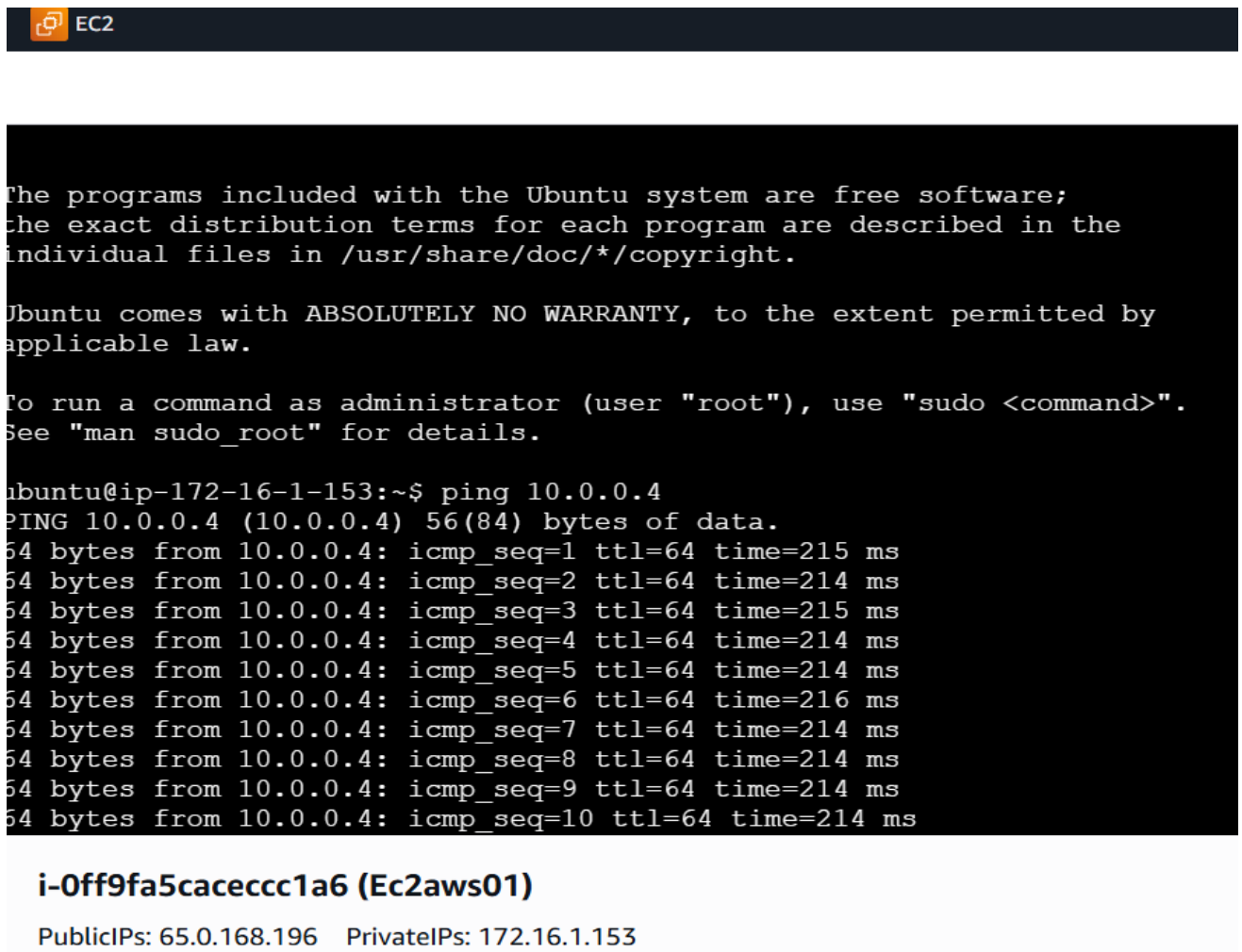
Next

[Download a template for automation](#)

- ✓ Next Go to AWS instances
- ✓ Select instance click on connect then opens Cloudshell.



- ✓ In Cloud shell type ping (enter Azure VM private IP)-check below



- ✓ In Azure to connect the VM we use **BASTION** (bash)